Henry Gao

(347) 440-6805 | gao.hen@northeastern.edu | linkedin.com/in/hengao | Portfolio: henrygao.vercel.app

EDUCATION

Northeastern University, Boston, MA *Expected May 2026*

Bachelor of Science in Computer Engineering *GPA: 3.51*

Coursework: Electronics, Circuits and Signals, Algorithms, Digital Design, Computer Architecture, Robotics, FPGA Design

Certifications & Skills: Certified SOLIDWORKS Associate | Circuit Design | Oscilloscope | Multimeter | Altium | Kicad | Python | C++ | LabVIEW | System Verilog | LTSpice | Arduino | Git | CAD | Linux

WORK EXPERIENCE

Bevi **Boston, MA**

Electrical Engineer Jul – Dec 2025

- Drove bring-up testing of key PCB boards, uncovering critical defects and enabling their successful deployment into production
- Designed and deployed an automated hardware validation system to characterize touchscreen response under varying capacitive loads, reducing manual testing by 80+ hours.
- Identified and fixed several bugs found within head unit control board current protection circuits

PHILIPS **Andover, MA**

Hardware Verification Support Engineer May – Dec 2024

- Spearheaded the design and implementation of a modular robotic test system to validate the operating limits of next-generation ECG and SPO2 monitors, saving Philips over 150+ hours per product in testing
- Built universal automation software to simplify operation of robotic systems, ensuring a more streamlined programming process for any future testing robots

PARSES Lab **Boston, MA**

Undergraduate Researcher Jan 2024 – Jan 2025

- Constructed flexible circuits for a wearable sleeve that detects if a patient has edema
- Collaborated on the design for stretch-sensitive capacitors used for sensing muscular hypertrophy

PROJECT & LEADERSHIP EXPERIENCE

Mira | Forge | Forge Product Development Studio | Project Lead Sep – Dec 2025

- Led a team of 6 engineers to design Mira, a smart mirror that self-cleans, blow dries your hair, and displays important day-to-day information
- Designed system level circuit for hair blowing, screen display, and self-cleaning using an ESP32 to control and communicate with a weather API and sensors via I2C, SPI, and UART protocols
- Developed fully modular hair blowing system using SOLIDWORKs air flow analysis tool

Greentower | Generate Product Development Studio | Electrical Engineer Sep – Dec 2025

- Created Greentower, a modular at home hydroponics system that automates watering, nutrient delivery, and sunlight exposure to growing plants
- Designed a custom pH sensing PCB board in Altium to read and filter analog voltage signals from a pH Sensor over SPI protocol
- Collaborated on the development of a main PCBA for the onboard power, MCU, motor, and LED systems

Medical Device Mate Test Robot | PHILIPS | Hardware Engineer May – Dec 2024

- Mimicked human behavior by wiggling to mate and un-mate ECG and SPO2 cables from corresponding monitors
- Utilizes C++ and LabVIEW to configure cycle speed, time, and force
- Successfully supported 35k test cycles while recording force values

CleanBoil | Forge Product Development Studio | Electrical Engineer Jan – Apr 2024

- Utilizes a silicon heater to heat up and boil water whenever the activation button is pressed
- Successfully filtered out 99% of sediments from unclean water with the usage of carbon filters
- Uses a relay to efficiently switch between two different batteries to regulate power efficiency

PUBLICATIONS

D. Leblebicioglu, H. Gao, I. Ampomah Mensah, M. S. Khan, and K. L. Dorsey, “Soft capacitive sensor and wearable sleeve towards measuring fluid retention,” in *Proc. IEEE SENSORS Conf.*, Kobe, Japan, 2024, pp. 1–4.